$(t_0\ \sigma_0\ Q_0\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{spawn}\ v_{coro})]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [spawn]
$(t_1\ \sigma_1\ Q_1\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F\ \ldots)\ FS_1\ \ldots))$

$(t_0\ \sigma_0\ Q_0\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{run}\ v)]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-run]
$(t_1\ \sigma_1\ Q_1\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where $(\text{ptr}\ x_{async}) = \text{malloc}[\![\sigma_0]\!]$

$(t_0\ \sigma_0\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{tagged}\ x_{running}\ E_{inner}[(\text{await}\ (\text{task}\ x_{async}))])]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [await-task-pending]
$(t_1\ \sigma_1\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{tag}\ x_{running})]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where (not in-tag?$[\![E_{inner}]\!]$), (not task-ready?$[\![\sigma_0, x_{async}]\!]$)

$(t_0\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-await-done]
$(t_1\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[v]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where $(\text{done}\ v) = \text{task-status}[\![v_{obj}]\!]$

$(t_0\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-await-cancelled]
$(t_1\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{struct})]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where $v_{obj} = \text{lookup}[\![\sigma, x_{async}]\!]$, $\text{cancelled} = \text{task-status}[\![v_{obj}]\!]$

$(t_0\ \sigma\ Q_0\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ v_{coro}\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-reschedule]
$(t_1\ \sigma\ Q_1\ (FS_0\ \ldots\ (\text{stack}\ F\ \ldots)\ FS_1\ \ldots))$
where (async? l), $x_{async} = l$, $v_{obj} = \text{lookup}[\![\sigma, x_{async}]\!]$, task-coro-eq?$[\![v_{obj}, v_{coro}]\!]$, $(\text{pending}\ \_\ \ldots) = \text{task-status}[\![v_{obj}]\!]$

$(t_0\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ v_{coro}\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-reschedule-cancelled]
$(t_1\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ F\ \ldots)\ FS_1\ \ldots))$
where (async? l), $x_{async} = l$, $v_{obj} = \text{lookup}[\![\sigma, x_{async}]\!]$, task-coro-eq?$[\![v_{obj}, v_{coro}]\!]$, $\text{cancelled} = \text{task-status}[\![v_{obj}]\!]$

$(t_0\ \sigma_0\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ v\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [task-return]
$(t_1\ \sigma_1\ Q\ (FS_0\ \ldots\ (\text{stack}\ F\ \ldots)\ FS_1\ \ldots))$
where (async? l), $x_{async} = l$, $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$, (not task-coro-eq?$[\![v_{obj}, v]\!]$)

$(t_0\ \sigma_0\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{cancel}\ (\text{task}\ x_{async}))]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [cancel]
$(t_1\ \sigma_1\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{void})]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$

$(t_0\ \sigma_0\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{os/io}\ natural\ v)]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [os/io]
$(t_1\ \sigma_1\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{spawn}\ (\text{tag}\ x_{tag}))]\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where $(x_{dummy}\ x_{tag}) = \text{lib:gensyms}[\![\sigma_0, \sigma_0]\!]$, $\sigma_1 = \text{ext1}[\![\sigma_0, (x_{tag}\ (\text{coroutine}\ (\text{lambda}\ (x_{dummy})$
$\text{lib:begin}[\![x_{dummy},$
$\text{lib:while}[\![(\text{<=}\ (\text{os/time})\ \text{lib:}\Sigma[\![t_0, natural]\!]),$
$(\text{yield}\ (\text{tag}\ x_{tag}))]\!],$
$v]\!])))]\!]$

$(t_0\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[(\text{os/time})]\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [os/time]
$(t_1\ \sigma\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ E[t_0]\ l)\ F\ \ldots)\ FS_1\ \ldots))$

$(t_0\ \sigma\ Q_0\ (FS_{main}\ FS_0\ \ldots\ (\text{stack})\ FS_1\ \ldots)) \longrightarrow$ [thread-work-steal]
$(t_1\ \sigma\ Q_1\ (FS_{main}\ FS_0\ \ldots\ (\text{stack}\ F_{head})\ FS_1\ \ldots))$
where $(F_{head}\ Q_1) = \text{q-pop}[\![Q_0]\!]$, all-busy?$[\![FS_0, \ldots]\!]$, $t_1 = \text{step}[\![t_0]\!]$

$(t_0\ \sigma_0\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ e_0\ l)\ F\ \ldots)\ FS_1\ \ldots)) \longrightarrow$ [base-step]
$(t_1\ \sigma_1\ Q\ (FS_0\ \ldots\ (\text{stack}\ (\text{frame}\ e_1\ l)\ F\ \ldots)\ FS_1\ \ldots))$
where (not value?$[\![e]\!]$), $(\sigma_1\ e_1) = \Downarrow\text{base}[\![\sigma_0, e_0]\!]$, $t_1 = \text{step}[\![t_0]\!]$

$(t_0\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{tag}\ x_{coro}))]\ l))\ FS_1\ \ldots)) \longrightarrow$ [block]
$(t_1\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{spawn}\ (\text{tag}\ x_{coro})))]\ l))\ FS_1\ \ldots))$

$(t_0\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{task}\ x_{async}))]\ l))\ FS_{rest}\ \ldots)) \longrightarrow$ [unblock]
$(t_1\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[v]\ l))\ FS_{rest}\ \ldots))$
where $(\text{done}\ v) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$